

Sahir Sood

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Education

Simon Fraser University

Expected Graduation: Aug 2027

Bachelor of Science in Computer Science and Finance (Combined Degree)

Burnaby, BC

- International Exchange Program — University of Leeds (Spring 2024)

Experience

Royal Bank of Canada

May 2026 – Present

Software Developer Intern

Vancouver, BC

- Designing an agentic employee discovery platform to identify relevant employees from natural-language skill queries.
- Built a skill-to-employee ranking pipeline across **5,000+** profiles, combining structured routing and LLM-based reasoning to surface relevant internal expertise.
- Migrated **2** internal products into standalone apps, decoupling deployment pipelines, enabling independent releases.
- Developed enterprise data explorer search, filtering, visualization, and export workflows across **25,000+** records, enabling analysts to investigate relationships and generate reports.
- Drove a governance lineage platform to production through deployment hardening and vulnerability remediation.

RedBrick (Paved)

Jan 2026 – Apr 2026

Software Developer Intern

Vancouver, BC

- Prevented cache stampedes in a **235M+** user system by coalescing concurrent requests and caching queries in Redis, reducing database load by **70%** and reducing latency.
- Built a distributed event ingestion pipeline (Next.js, Ruby) preventing duplicate processing under concurrency using idempotency keys and cross-pipeline validation.
- Redesigned targeting using array-based PostgreSQL and legacy data backfills, reducing bid mismatches by **25%**.
- Owned development of a Rails-based newsletter conversion system end-to-end and presented architecture and technical tradeoffs to leadership.

Mothertongue

May 2025 – Jul 2025

Software Developer

Vancouver, BC

- Ensured schema-valid GPT-4o outputs under concurrent usage by enforcing schema-constrained JSON generation, runtime validation, and request deduplication in an Express API.
- Reduced API costs by implementing rolling context windows, request throttling, and duplicate-call suppression, lowering monthly spend by **~78%**.
- Deployed a Next.js orchestration layer to sync events with backend services, ensuring consistent state across tabs.

Projects

Agentic Market Simulation Platform | [Live Demo](#) | [GitHub](#) | C++, Python, FastAPI, React, PostgreSQL, MCP

- Built a **C++** limit-order-book engine with price-time priority, partial fills, and O(1) cancellation via indexed order lookups; benchmarked at **800K+** orders/sec.
- Orchestrated **10** Claude/OpenAI agents across **5** personas, benchmarking under deterministic risk controls.
- Built a **RAG** pipeline over 10-K, 10-Q, and 8-K filings plus market news, using chunking, embeddings, metadata-filtered retrieval, and citations to ground trading decisions.
- Exposed market, evidence, and risk context through **5 MCP** tools; orders are re-validated before execution.

Spotify Playlist Generator | [GitHub](#) | Python, Node.js, React, AWS, PostgreSQL, Redis

- Designed a 4-Lambda pipeline with EventBridge scheduling, processing **1000+** tracks per user in DynamoDB.
- Implemented a Python Lambda that invoked GPT-4o with caching to reduce redundant API calls and generated 10 user taste profiles via K-Means clustering.
- Deployed a Node.js Lambda handling Spotify OAuth and playlist creation at **~200** ms via GitHub Actions.

Technical Skills

Languages: Python, C++, TypeScript, JavaScript, SQL, Ruby, Java

Backend & AI: RAG, MCP, LLM Agents, FastAPI, Node.js, REST APIs, WebSockets

Databases & Cloud: PostgreSQL, Redis, DynamoDB, MongoDB, AWS (Lambda, EventBridge), Docker

Frontend & Tooling: React, Next.js, Ruby on Rails, GitHub Actions, Jest, Selenium, Postman